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Single-photon detection device, single-photon detector, and single-photon detector array

Abstract

A single-photon detection device includes a first well having a first conductivity type, a second well provided on the first well and having a second conductivity type that is different from the first conductivity type, a first depletion forming region provided on the second well and having the first conductivity type, a main depletion region provided between the first well and the second well, and a first sub-depletion region provided between the second well and the first depletion forming region, wherein the first well and the first depletion forming region are spaced apart from each other by the second well.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9209320	12/2014	Webster	N/A	N/A
9299732	12/2015	Webster et al.	N/A	N/A
9312401	12/2015	Webster	N/A	N/A
9728667	12/2016	Johnson et al.	N/A	N/A
10438987	12/2018	Mandai et al.	N/A	N/A
2009/0315135	12/2008	Finkelstein	257/E31.113	H01L 31/107
2010/0013458	12/2009	Lany	324/123R	G01R 19/0092
2010/0019295	12/2009	Henderson et al.	N/A	N/A
2012/0261729	12/2011	Finkelstein et al.	N/A	N/A
2015/0054111	12/2014	Niclass	257/438	H01L 27/1446
2015/0200314	12/2014	Webster	N/A	N/A
2015/0340391	12/2014	Webster	N/A	N/A
2016/0099371	12/2015	Webster	N/A	N/A
2017/0186798	12/2016	Yang	N/A	H01L 27/14632
2020/0091208	12/2019	Otake	N/A	H01L 27/1463
2021/0028202	12/2020	Inoue	N/A	H01L 27/14643
2021/0126025	12/2020	Kennedy	N/A	H01L 27/1464
2022/0262970	12/2021	Murakami	N/A	H01L 31/035272

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2016041746	12/2014	JP	N/A
101666440	12/2015	KR	N/A
1020190049598	12/2018	KR	N/A
1020200110717	12/2019	KR	N/A

OTHER PUBLICATIONS

I. Oshiyama et al., “Near-infrared Sensitivity Enhancement of a Back-illuminated Complementary Metal Oxide Semiconductor Image Sensor with a Pyramid Surface for Diffraction Structure”, 2017 IEEE International Electron Devices Meeting (IEDM), Dec. 2017, pp. 16.4.1 ~ 16.4.4. cited by applicant

Isamu Takai et al., “Single-Photon Avalanche diode with Enhanced NIR-Sensitivity for Automotive LIDAR Systems”, Sensors, 16, 459, pp. 1-9, Mar. 30, 2016. cited by applicant

Kai Zang et al., “Enhanced silicon single-photon avalanche diode based on light trapping”, 2018 Conference on Lasers and Electro-Optics (CLEO), May 13-18, 2018. cited by applicant

Kai Zang et al., “Silicon single-photon avalanche diodes with nano-structured light trapping”, Nature Communications, 3:628, Sep. 20, 2017, pp. 1-6. cited by applicant

Ceccarelli, Francesco, et al; “Recent Advances and Future Perspectives of Single-Photon Avalanche Diodes for Quantum Photonics Applications”, Advanced Quantum Technologies 4.2 (first published Dec. 2020): <https://doi.org/10.1002/qute.202000102>. cited by applicant

International Search Report and Written Opinion of PCT/KR2022/012002 mailed on Nov. 28, 2022. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0115811, filed on Aug. 31, 2021, and Korean Patent Application No. 10-2021-0137638, filed on Oct. 15, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) One or more embodiments relate to a single-photon detection device, a single-photon detector, and a single-photon detector array.

2. Description of the Related Art

(3) An avalanche photodiode (APD) is a solid-state photodetector in which a high bias voltage is applied to a p-n junction to provide a high first stage gain due to avalanche multiplication. When an incident photon with sufficient energy to emit an electron reaches a photodiode, an electron-hole pair (EHP) is generated. A high electric field rapidly accelerates photo-generated electrons toward (+), additional electron-hole pairs are sequentially generated due to impact ionization caused by the accelerated electrons, and then all of the electrons are accelerated toward the (+) node. Likewise, holes are rapidly accelerated toward (−), and the same phenomenon occurs. As this process repeats, avalanche multiplication of photo-generated carriers and an output current occurs. Accordingly, an APD is a semiconductor-based device that operates in a similar manner to that of a photomultiplier tube. A linear-mode APD is an effective amplifier in which a gain may be set by controlling a bias voltage and a gain of tens to thousands may be achieved in a linear mode.

(4) A single-photon avalanche diode (SPAD) is an APD in which a p-n junction is biased above its breakdown voltage to operate in a Geiger mode, so that a single incident photon may trigger an avalanche to generate very large current and thus an easily measurable current pulse may be

obtained along with a quenching resistor or circuit. That is, the SPAD operates as a device that generates a large current pulse when compared to a linear-mode APD in which a gain may not be sufficient at a low light intensity. After the avalanche is triggered, the quenching resistor or circuit is used to reduce a bias below the breakdown voltage in order to quench an avalanche process. Once the avalanche process is quenched, the bias voltage is raised again above the breakdown voltage to reset the SPAD for detection of another photon (i.e., known as recharge the SPAD). (5) An APD or a SPAD may have electrons generated due to surface defects. The electrons generated due to the surface defects may be introduced into the APD or the SPAD, and may be multiplied. Accordingly, a noise signal may be generated.

SUMMARY

- (6) One or more embodiments include a photo-detection device having low noise.
- (7) One or more embodiments include a photodetector having low noise.
- (8) However, the embodiments are examples, and do not limit the scope of the disclosure.
- (9) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.
- (10) According to one or more embodiments, a single-photon detection device includes a first well having a first conductivity type, a second well provided on the first well and having a second conductivity type that is different from the first conductivity type, a first depletion forming region provided on the second well and having the first conductivity type, a main depletion region provided between the first well and the second well, and a first sub-depletion region provided between the second well and the first depletion forming region, wherein the first well and the first depletion forming region are spaced apart from each other by the second well.
- (11) The single-photon detection device may further include a second depletion forming region provided opposite to the second well with the first well therebetween, and a second sub-depletion region provided between the second depletion forming region and the first well, wherein the second depletion forming region has the second conductivity type.
- (12) The single-photon detection device may further include a first contact region electrically connected to the first well, wherein the first contact region is spaced apart from the second well and has the first conductivity type, wherein a doping concentration of the first contact region is higher than a doping concentration of the first well.
- (13) The single-photon detection device may further include a relaxation region provided between the first contact region and the first well, wherein the relaxation region has the first conductivity type, wherein a doping concentration of the relaxation region is lower than a doping concentration of the first contact region.
- (14) The single-photon detection device may further include a second contact region electrically connected to the second well, wherein the second contact region has the second conductivity type, wherein a doping concentration of the second contact region is higher than a doping concentration of the second well.
- (15) The second contact region and the first depletion forming region may contact each other.
- (16) The second contact region and the first depletion forming region may be spaced apart from each other.
- (17) The single-photon detection device may further include a guard ring region provided on a side surface of the second well, wherein the guard ring region has the second conductivity type.
- (18) The guard ring region may surround the second well.
- (19) The first well may extend from a bottom surface of the second well to a side surface of the second well.
- (20) The single-photon detection device may further include a third well surrounding the first well and the second well, wherein the third well has the first conductivity type.
- (21) The single-photon detection device may further include a second depletion forming region provided opposite to the first well with the third well therebetween, and a third sub-depletion

region provided between the second depletion forming region and the third well, wherein the second depletion forming region has the second conductivity type.

(22) The single-photon detection device may further include a third contact region provided on the third well and spaced apart from the second well, wherein the third contact region has the first conductivity type.

(23) A doping concentration of the third well may be lower than a doping concentration of the first well.

(24) The third well may contact a side surface of the second well.

(25) According to one or more embodiments, a single-photon detector includes a single-photon detection device configured to generate an electrical signal by receiving light, a control layer configured to process the electrical signal, and a lens unit configured to receive the light from outside and focus the light on the single-photon detection device, wherein the single-photon detection device includes a first well having a first conductivity type, a second well provided on the first well and having a second conductivity type that is different from the first conductivity type, a first depletion forming region provided on the second well and having the first conductivity type, a main depletion region provided between the first well and the second well, and a first sub-depletion region provided between the second well and the first depletion forming region, wherein the first well and the first depletion forming region are spaced apart from each other by the second well.

(26) The control layer may be disposed between the single-photon detection device and the lens unit.

(27) The single-photon detection device may be disposed between the control layer and the lens unit.

(28) The first depletion forming region may be disposed closer to the control layer than the lens unit.

(29) The control layer may include an insulating layer, wirings, and circuits, wherein the wirings electrically connect the first well and the second well to the circuits, and the circuits include a quenching circuit and a readout circuit.

(30) The circuits may further include at least one of a recharge circuit, a memory, an amplification circuit, a counter, and a gate circuit.

(31) The single-photon detector may include at least one of a color filter, a bandpass filter, an anti-reflection coating, a two-dimensional (2D) nanomaterial, and an organic material provided on the single-photon detection device.

(32) According to one or more embodiments, a single-photon detector array includes a plurality of single-photon detection devices configured to generate an electrical signal by receiving light, a control layer configured to process the electrical signal, and a plurality of lenses configured to receive the light from outside and respectively focus the light on the plurality of single-photon detection devices, wherein each of the plurality of single-photon detection devices includes a first well having a first conductivity type, a second well provided on the first well and having a second conductivity type that is different from the first conductivity type, a first depletion forming region provided on the second well and having the first conductivity type, a main depletion region provided between the first well and the second well, and a first sub-depletion region provided between the second well and the first depletion forming region, wherein the first well and the first depletion forming region are spaced apart from each other by the second well.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of certain embodiments will be more apparent from the following description taken in conjunction with the accompanying drawings, in

which:

- (2) FIG. 1 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment;
- (3) FIG. 2 is a plan view illustrating the single-photon detection device of FIG. 1, according to an embodiment;
- (4) FIG. 3 is a plan view illustrating the single-photon detection device of FIG. 1, according to an embodiment;
- (5) FIG. 4 is a plan view illustrating the single-photon detection device of FIG. 1, according to an embodiment;
- (6) FIG. 5 is a plan view illustrating the single-photon detection device of FIG. 1, according to an embodiment;
- (7) FIG. 6 is a plan view illustrating the single-photon detection device of FIG. 1, according to an embodiment;
- (8) FIG. 7 is a plan view illustrating the single-photon detection device of FIG. 1, according to an embodiment;
- (9) FIG. 8 is a plan view illustrating the single-photon detection device of FIG. 1, according to an embodiment;
- (10) FIG. 9 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment;
- (11) FIG. 10 is a plan view illustrating the single-photon detection device of FIG. 9, according to an embodiment;
- (12) FIG. 11 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment;
- (13) FIG. 12 is a plan view illustrating the single-photon detection device of FIG. 11, according to an embodiment;
- (14) FIG. 13 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment;
- (15) FIG. 14 is a plan view illustrating the single-photon detection device of FIG. 13, according to an embodiment;
- (16) FIG. 15 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment;
- (17) FIG. 16 is a plan view illustrating the single-photon detection device of FIG. 15, according to an embodiment;
- (18) FIG. 17 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment;
- (19) FIG. 18 is a cross-sectional view illustrating a single-photon detector, according to an embodiment;
- (20) FIG. 19 is a cross-sectional view illustrating a single-photon detector, according to an embodiment;
- (21) FIG. 20 is a plan view illustrating a single-photon detector array, according to an embodiment;
- (22) FIG. 21 is a cross-sectional view taken along line I-I' of FIG. 20;
- (23) FIG. 22 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment;
- (24) FIG. 23 is a plan view illustrating the single-photon detection device of FIG. 22, according to an embodiment;
- (25) FIG. 24 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment;
- (26) FIG. 25 is a plan view illustrating the single-photon detection device of FIG. 24, according to an embodiment;
- (27) FIG. 26 is a cross-sectional view illustrating a single-photon detection device, according to an

embodiment;

(28) FIG. 27 is a plan view illustrating the single-photon detection device of FIG. 26, according to an embodiment;

(29) FIG. 28 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment;

(30) FIG. 29 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment;

(31) FIG. 30 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment;

(32) FIG. 31 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment;

(33) FIG. 32 is a block diagram for describing an electronic device, according to an embodiment; and

(34) FIGS. 33 and 34 are conceptual views illustrating a case where a light detection and ranging (LiDAR) device is applied to a vehicle, according to an embodiment.

DETAILED DESCRIPTION

(35) Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of the present description. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

(36) Hereinafter, embodiments of the disclosure will be described with reference to the drawings. Like reference numerals denote like elements throughout, and in the drawings, sizes of elements may be exaggerated for clarity and convenience of explanation. Also, the embodiments described below are merely examples, and various modifications may be made from the embodiments.

(37) When an element is referred to as being “on” another element, it may be directly on the other element, or intervening elements may be present therebetween.

(38) The singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. When a portion “includes” an element, another element may be further included, rather than excluding the existence of the other element, unless otherwise described.

(39) Also, the term such as “. . . unit” used therein refers to a unit for processing at least one function or operation.

(40) FIG. 1 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment.

(41) Referring to FIG. 1, a single-photon detection device 1 may be provided. The single-photon detection device 1 may include a single-photon avalanche diode (SPAD). For example, the SPAD may be referred to as a Geiger-mode avalanche photodiode (APD) (G-APD). The single-photon detection device 1 may include a first well 110, a second well 120, a first contact region 114, a first relaxation region 112, a first depletion forming region 130, a second contact region 122, and a second depletion forming region 100. The first well 110, the second well 120, the first contact region 114, the first relaxation region 112, the first depletion forming region 130, the second contact region 122, and the second depletion forming region 100 may be formed by doping a semiconductor substrate (e.g., a p-type or n-type silicon (Si) substrate). For example, the semiconductor substrate may be a p-type or n-type epitaxial (epi) layer. The second depletion forming region 100 may be formed by using an epitaxial growth process.

(42) The first well 110 may have a first conductivity type. When a conductivity type of the

semiconductor substrate is p-type, a conductivity type of the first well **110** may be n-type. When a conductivity type of the first well **110** is n-type, the first well **110** may be formed by implanting a group 5 element (e.g., phosphorus (P), arsenic (As), or antimony (Sb)), a group 6 element, or a group 7 element as impurities into a silicon (Si) substrate. When a conductivity type of the semiconductor substrate is n-type, a conductivity type of the first well **110** may be p-type. When a conductivity type of the first well **110** is p-type, the first well **110** may be formed by implanting a group 3 element (e.g., boron (B), aluminum (Al), gallium (Ga), or indium (In)) or a group 2 element as impurities into a silicon (Si) substrate. For example, a doping concentration of the first well **110** may range from about $1 \times 10^{14} \text{ cm}^{-3}$ to about $1 \times 10^{19} \text{ cm}^{-3}$. In an example, a doping concentration of the first well **110** may have a gradient that decreases toward an upper portion of the first well **110**. In another example, the first well **110** may have a uniform doping concentration.

(43) The second well **120** may have a second conductivity type that is different from the first conductivity type. When a conductivity type of the first well **110** is n-type, a conductivity type of the second well **120** may be p-type. When a conductivity type of the first well **110** is p-type, a conductivity type of the second well **120** may be n-type. For example, a doping concentration of the second well **120** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{18} \text{ cm}^{-3}$.

(44) A first main depletion region **30** may be formed in a portion adjacent to an interface between the first well **110** and the second well **120**. When a reverse bias is applied to the single-photon detection device **1**, a strong electric field may be applied to the first main depletion region **30**. For example, when the single-photon detection device **1** operates as an SPAD, a maximum strength of the electric field may range from about $1 \times 10^5 \text{ V/cm}$ to about $1 \times 10^6 \text{ V/cm}$. Because electrons may be multiplied by an electric field of the first main depletion region **30**, the first main depletion region **30** may be referred to as a multiplication region.

(45) The first contact region **114** may be located on the first well **110**. The first contact region **114** may be electrically connected to a circuit outside the single-photon detection device **1**. For example, a voltage may be applied to the first well **110** from the circuit outside the single-photon detection device **1** through the first contact region **114**. The first contact region **114** may be located on a side surface of the second well **120**. The first contact region **114** may surround the second well **120**. For example, the first contact region **114** may have an annular shape extending along the side surface of the second well **120**. The first contact region **114** may be spaced apart from the second well **120**. The first well **110** may extend to a portion between the first contact region **114** and the second well **120**. For example, the portion between the first contact region **114** and the second well **120** may be filled with the first well **110**.

(46) The first contact region **114** may have the first conductivity type. A doping concentration of the first contact region **114** may be higher than a doping concentration of the first well **110**. For example, a doping concentration of the first contact region **114** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{22} \text{ cm}^{-3}$.

(47) The first relaxation region **112** may be located or provided between the first contact region **114** and the first well **110**. The first relaxation region **112** may be electrically connected to the first contact region **114** and the first well **110**. The first relaxation region **112** may reduce a difference between a bandgap of the first contact region **114** and a bandgap of the first well **110**. The first relaxation region **112** may be located or provided on a side surface of the second well **120**. The first relaxation region **112** may surround the second well **120**. For example, the first relaxation region **112** may have an annular shape extending in a direction in which the side surface of the second well **120** extends. The first relaxation region **112** may be located or provided under the first contact region **114**. For example, side surfaces of the first relaxation region **112** may be coplanar with side surfaces of the first contact region **114**. The first relaxation region **112** may be spaced apart from the second well **120**. The first well **110** may extend into a portion between the first relaxation

region **112** and the second well **120**. For example, the portion between the first relaxation region **112** and the second well **120** may be filled with the first well **110**. The first relaxation region **112** may be formed to be deeper than the second well **120**. For example, a bottom surface of the first relaxation region **112** may be spaced apart farther away from a top surface of the single-photon detection device **1** than a bottom surface of the second well **120**. The first relaxation region **112** may have the first conductivity type. A doping concentration of the first relaxation region **112** may be lower than a doping concentration of the first contact region **114**, and may be higher than a doping concentration of the first well **110**. For example, a doping concentration of the first relaxation region **112** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{19} \text{ cm}^{-3}$.

(48) The first depletion forming region **130** may be located or provided on the second well **120**. The first depletion forming region **130** may have the first conductivity type. A doping concentration of the first depletion forming region **130** may be higher than a doping concentration of the first well **110**. For example, a doping concentration of the first depletion forming region **130** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{22} \text{ cm}^{-3}$.

(49) A first sub-depletion region **10** may be formed in a portion adjacent to an interface between the first depletion forming region **130** and the second well **120**. The first sub-depletion region **10** may reduce or substantially prevent electrons or holes other than electron-hole pairs generated by photons in the single-photon detection device **1** from being provided to a first main depletion region **30**. For example, electrons or holes other than electron-hole pairs generated by photons in the single-photon detection device **1** may be generated due to defects of a surface of the single-photon detection device **1** adjacent to the first sub-depletion region **10**. The first sub-depletion region **10** may reduce or substantially prevent the electrons or holes due to the surface defects of the single-photon detection device **1** from moving to the first main depletion region **30**.

(50) The second contact region **122** may be located or provided on the second well **120**. The second contact region **122** may be electrically connected to a circuit outside the single-photon detection device **1**. For example, when the single-photon detection device **1** is a SPAD, the single-photon detection device **1** may be electrically connected to a quenching resistor or circuit and other pixel circuits through the second contact region **122**. The quenching resistor or circuit may stop an avalanche effect and may enable the SPAD to detect another photon. The other pixel circuits may include a reset or recharge circuit, a memory, an amplification circuit, a counter, and a gate circuit, and may transmit signal current to the single-photon detection device **1** or may receive signal current from the single-photon detection device **1**. The second contact region **122** may be located or provided on a side surface of the first depletion forming region **130**. The second contact region **122** may surround the first depletion forming region **130**. For example, the second contact region **122** may have an annular shape extending along the side surface of the first depletion forming region **130**. The second contact region **122** may have an inner surface adjacent to the first depletion forming region **130** and an outer surface located or provided opposite to the inner surface. In an embodiment, the second contact region **122** may contact the first depletion forming region **130**. For example, the inner surface of the second contact region **122** may contact the side surface of the first depletion forming region **130**. In an embodiment, the second contact region **122** may be spaced apart from the first depletion forming region **130**. For example, the inner surface of the second contact region **122** may face the first depletion forming region **130** and may be spaced apart from the first depletion forming region **130**. The second contact region **122** may have a second conductivity type. A doping concentration of the second contact region **122** may be higher than a doping concentration of the second well **120**. For example, a doping concentration of the second contact region **122** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{22} \text{ cm}^{-3}$. The outer surface of the second contact region **122** may be covered by the second well **120**. In other words, the second well **120** may extend to the outer surface of the second contact region **122**.

(51) The second depletion forming region **100** may be located or provided opposite to the second

well **120** with the first well **110** therebetween. The second depletion forming region **100** may have the second conductivity type. For example, a doping concentration of the second depletion forming region **100** may range from about $1 \times 10^{14} \text{ cm}^{-3}$ to about $1 \times 10^{22} \text{ cm}^{-3}$.

(52) A second sub-depletion region **20** may be formed in a portion adjacent to an interface between the second depletion forming region **100** and the first well **110**. The second sub-depletion region **20** may reduce or substantially prevent electrons or holes other than electron-hole pairs generated by photons in the single-photon detection device **1** from being provided to the first main depletion region **30**. For example, electrons or holes other than electron-hole pairs generated by photons in the single-photon detection device **1** may be generated due to surface defects of the single-photon detection device **1** adjacent to the second sub-depletion region **20**. The second sub-depletion region **20** may reduce or substantially prevent the electrons or holes due to the surface defects of the single-photon detection device **1** from moving to the first main depletion region **30**.

(53) The single-photon detection device **1** may have surface defects. Electrons or holes may be generated due to the surface defects of the single-photon detection device **1**. When the electrons or holes generated due to the surface defects move to the first main depletion region **30**, the electrons or holes may be multiplied and may generate a noise signal. That is, even when a photon is not provided to the single-photon detection device **1**, the single-photon detection device **1** may generate a noise signal.

(54) The first sub-depletion region **10** and the second sub-depletion region **20** of the disclosure may reduce or substantially prevent electrons or holes generated due to surface defects of the single-photon detection device **1** from moving to the first main depletion region **30**. Accordingly, the single-photon detection device **1** having low noise may be provided. Both or only one of the first sub-depletion region **10** and the second sub-depletion region **20** may be implemented.

(55) FIG. 2 is a plan view illustrating the single-photon detection device of FIG. 1, according to an embodiment.

(56) Referring to FIG. 2, a single-photon detection device **1a(1)** may have a circular shape. In detail, the first depletion forming region **130** may have a circular shape, and the second contact region **122**, the first contact region **114**, the second well **120**, and the first well **110** may each have a circular annular shape. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may surround the first depletion forming region **130**. For example, the first depletion forming region **130**, the second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may have the same center. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may be sequentially arranged away from the first depletion forming region **130**.

(57) FIG. 3 is a plan view illustrating the single-photon detection device of FIG. 1, according to an embodiment.

(58) Referring to FIG. 3, a single-photon detection device **1b(1)** may have a square shape. In detail, the first depletion forming region **130** may have a square shape, and the second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may each have a square annular shape. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may surround the first depletion forming region **130**. For example, the first depletion forming region **130**, the second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may have the same center. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may be sequentially arranged away from the first depletion forming region **130**.

(59) FIG. 4 is a plan view illustrating the single-photon detection device of FIG. 1, according to an embodiment.

(60) Referring to FIG. 4, a single-photon detection device **1c(1)** may have a square shape with round corners. In detail, the first depletion forming region **130** may have a square shape with round corners, and the second contact region **122**, the second well **120**, the first well **110**, and the first

contact region **114** may each have a square annular shape with round corners. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may surround the first depletion forming region **130**. For example, the first depletion forming region **130**, the second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may have the same center. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may be sequentially arranged away from the first depletion forming region **130**.

(61) FIG. **5** is a plan view illustrating the single-photon detection device of FIG. **1**, according to an embodiment.

(62) Referring to FIG. **5**, a single-photon detection device **1d(1)** may have a rectangular shape. In detail, the first depletion forming region **130** may have a rectangular shape, and the second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may each have a rectangular annular shape. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may surround the first depletion forming region **130**. For example, the first depletion forming region **130**, the second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may have the same center. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may be sequentially arranged away from the first depletion forming region **130**.

(63) FIG. **6** is a plan view illustrating the single-photon detection device of FIG. **1**, according to an embodiment.

(64) Referring to FIG. **6**, a single-photon detection device **1e(1)** may have a rectangular shape with round corners. In detail, the first depletion forming region **130** may have a rectangular shape with round corners, and the second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may each have a rectangular annular shape with round corners. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may surround the first depletion forming region **130**. For example, the first depletion forming region **130**, the second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may have the same center. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may be sequentially arranged away from the first depletion forming region **130**.

(65) FIG. **7** is a plan view illustrating the single-photon detection device of FIG. **1**, according to an embodiment.

(66) Referring to FIG. **7**, a single-photon detection device **1f(1)** may have an elliptical shape. In detail, the first depletion forming region **130** may have an elliptical shape, and the second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may each have an elliptical shape. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may surround the first depletion forming region **130**. For example, the first depletion forming region **130**, the second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may have the same center. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may be sequentially arranged away from the first depletion forming region **130**.

(67) FIG. **8** is a plan view illustrating the single-photon detection device of FIG. **1**, according to an embodiment.

(68) Referring to FIG. **8**, a single-photon detection device **1g(1)** may have an octagonal shape. In detail, the first depletion forming region **130** may have an octagonal shape, and the second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may each have an octagonal shape. The second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may surround the first depletion forming region **130**. For example, the first depletion forming region **130**, the second contact region **122**, the second well **120**, the first well **110**, and the first contact region **114** may have the same center. The second contact region **122**, the

second well **120**, the first well **110**, and the first contact region **114** may be sequentially arranged away from the first depletion forming region **130**.

(69) FIG. **9** is a cross-sectional view illustrating a single-photon detection device, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **1** will be omitted.

(70) Referring to FIG. **9**, a single-photon detection device **2** may be provided. The single-photon detection device **2** may include a SPAD. The single-photon detection device **2** may be substantially the same as the single-photon detection device **1** of FIG. **1**, except for shapes of the first contact region **114** and the first relaxation region **112**.

(71) Unlike in FIG. **1**, a width of the first contact region **114** may be less than a width of the first relaxation region **112**. The first contact region **114** may be located inside the first relaxation region **112**. Side surfaces of the first contact region **114** may be spaced apart from an outer surface of the first relaxation region **112**.

(72) A top surface of the first contact region **114** may be coplanar with a top surface of the first relaxation region **112**. For example, a top surface of the first contact region **114** may be located between top surfaces of the first relaxation region **112**.

(73) FIG. **10** is a plan view illustrating the single-photon detection device of FIG. **9**, according to an embodiment.

(74) Referring to FIG. **10**, the single-photon detection device **2** may have a circular shape. In detail, the first depletion forming region **130** may have a circular shape, and the second contact region **122**, the second well **120**, the first well **110**, the first relaxation region **112**, and the first contact region **114** may each have a circular annular shape. The second contact region **122**, the second well **120**, the first well **110**, the first relaxation region **112**, and the first contact region **114** may surround the first depletion forming region **130**. For example, the first depletion forming region **130**, the second contact region **122**, the second well **120**, the first well **110**, the first relaxation region **112**, and the first contact region **114** may have the same center. The second contact region **122**, the second well **120**, the first well **110**, the first relaxation region **112**, and the first contact region **114** may be sequentially arranged away from the first depletion forming region **130**.

(75) Although the single-photon detection device **2** has a circular shape, the disclosure is not limited thereto. In another example, as shown in FIGS. **3** through **8**, the single-photon detection device **2** may have a square shape, an octagonal shape, a square shape with round corners, a rectangular shape, a rectangular shape with round corners, or an elliptical shape.

(76) FIG. **11** is a cross-sectional view illustrating a single-photon detection device, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **1** will be omitted.

(77) Referring to FIG. **11**, a single-photon detection device **3** may be provided. The single-photon detection device **3** may include a SPAD. The single-photon detection device **3** may include the first well **110**, the second well **120**, the first contact region **114**, the first relaxation region **112**, the first depletion forming region **130**, the second contact region **122**, the second depletion forming region **100**, and a guard ring region **500**. The first well **110**, the second well **120**, the first contact region **114**, the first relaxation region **112**, the first depletion forming region **130**, the second contact region **122**, the second depletion forming region **100**, and the guard ring region **500** may be formed on a semiconductor substrate (e.g., a p-type or n-type silicon (Si) substrate). For example, the semiconductor substrate may include a p-type or n-type epitaxial layer. The second depletion forming region **100** may be formed by using an epitaxial growth method.

(78) The first well **110**, the second well **120**, the first contact region **114**, the first relaxation region **112**, the first depletion forming region **130**, the second contact region **122**, and the second depletion forming region **100** may be respectively substantially the same as the first well **110**, the second well **120**, the first contact region **114**, the first relaxation region **112**, the first depletion forming region **130**, the second contact region **122**, and the second depletion forming region **100** of FIG. **1**.

(79) The guard ring region **500** may be located or provided on a side surface of the second well **120**. The guard ring region **500** may prevent premature breakdown by reducing the concentration of an electric field at an edge of the second well **120**. Premature breakdown, which is a phenomenon where breakdown first occurs at an edge of the second well **120** before an electric field of a sufficient magnitude is applied to the first main depletion region **30**, occurs as an electric field is concentrated at the edge of the second well **120**. The guard ring region **500** may improve breakdown characteristics of the single-photon detection device **3**. The guard ring region **500** may surround the second well **120**. For example, the guard ring region **500** may have an annular shape extending along the side surface of the second well **120**. The guard ring region **500** may directly contact the second well **120**. The guard ring region **500** may be spaced apart from the first contact region **114** and the first relaxation region **112**. The first well **110** may extend to a portion between the guard ring region **500** and the first relaxation region **112** and a portion between the guard ring region **500** and the first contact region **114**. For example, the portion between the guard ring region **500** and the first relaxation region **112** and the portion between the guard ring region **500** and the first contact region **114** may be filled with the first well **110**. The guard ring region **500** may protrude from an interface between the second well **120** and the first well **110** to the first well **110**. The guard ring region **500** is not limited to being spaced apart from the first contact region **114** and the first relaxation region **112**. In another example, the guard ring region **500** may directly contact the first contact region **114** and the first relaxation region **112**. The guard ring region **500** may have the second conductivity type. A doping concentration of the guard ring region **500** may be lower than a doping concentration of the second well **120**. For example, a doping concentration of the guard ring region **500** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{18} \text{ cm}^{-3}$.

(80) According to the disclosure, the single-photon detection device **3** having low noise and improved breakdown characteristics may be provided.

(81) FIG. **12** is a plan view illustrating the single-photon detection device of FIG. **11**, according to an embodiment.

(82) Referring to FIG. **12**, the single-photon detection device **3** may have a circular shape. In detail, the first depletion forming region **130** may have a circular shape, and the second contact region **122**, the second well **120**, the guard ring region **500**, the first well **110**, and the first contact region **114** may each have a circular annular shape. The second contact region **122**, the second well **120**, the guard ring region **500**, the first well **110**, and the first contact region **114** may surround the first depletion forming region **130**. For example, the first depletion forming region **130**, the second contact region **122**, the second well **120**, the guard ring region **500**, the first well **110**, and the first contact region **114** may have the same center. The second contact region **122**, the second well **120**, the guard ring region **500**, the first well **110**, and the first contact region **114** may be arranged away from the first depletion forming region **130**.

(83) Although the single-photon detection device **3** has a circular shape, the disclosure is not limited thereto. In another example, as shown in FIGS. **3** through **8**, the single-photon detection device **3** may have a square shape, an octagonal shape, a square shape with round corners, a rectangular shape, a rectangular shape with round corners, or an elliptical shape.

(84) FIG. **13** is a cross-sectional view illustrating a single-photon detection device, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **11** will be omitted.

(85) Referring to FIG. **13**, a single-photon detection device **4** may be provided. The single-photon detection device **4** may be substantially the same as the single-photon detection device **3** of FIG. **11**, except for the first depletion forming region **130**, the second contact region **122**, and the guard ring region **500**.

(86) The second contact region **122** may be located or disposed on the guard ring region **500**. A side surface of the second contact region **122** may be coplanar with a side surface of the guard ring

region **500** adjacent to the second contact region **122**. Although the entire second contact region **122** is located or disposed on the guard ring region **500**, the disclosure is not limited thereto. In another example, a part of the second contact region **122** may be located or disposed on the guard ring region **500**. In another example, a width of the second contact region **122** may be less than a width of the guard ring region **500**. The second contact region **122** may be located inside the guard ring region **500**. Side surfaces of the second contact region **122** may be spaced apart from an outer surface of the guard ring region **500**.

(87) The second contact region **122** may be spaced apart from the first depletion forming region **130**. A part of the second well **120** may be located or disposed between the second contact region **122** and the first depletion forming region **130**. However, this is merely an example. In another example, the first depletion forming region **130** may extend to contact the second contact region **122** located or disposed on the guard ring region **500**.

(88) As the second contact region **122** is located or disposed on the guard ring region **500**, the first depletion forming region **130** may be provided in a larger area as that described with reference to FIG. **11**. Accordingly, the first sub-depletion region **10** may be formed wider than that described with reference to FIG. **11**.

(89) According to the disclosure, the single-photon detection device **4** having low noise and improved breakdown characteristics may be provided.

(90) FIG. **14** is a plan view illustrating the single-photon detection device of FIG. **13**, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **12** will be omitted.

(91) Referring to FIG. **14**, the single-photon detection device **4** may have a circular shape. Unlike in FIG. **12**, the guard ring region **500** (see FIG. **13**) may be covered by the second contact region **122**.

(92) Although the single-photon detection device **4** has a circular shape, the disclosure is not limited thereto. In another example, as shown in FIGS. **3** through **8**, the single-photon detection device **4** may have a square shape, an octagonal shape, a square shape with round corners, a rectangular shape, a rectangular shape with round corners, or an elliptical shape.

(93) FIG. **15** is a cross-sectional view illustrating a single-photon detection device, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **1** will be omitted.

(94) Referring to FIG. **15**, a single-photon detection device **5** may be provided. The single-photon detection device **5** may include a SPAD. The single-photon detection device **5** may include a third well **210**, a fourth well **220**, a fifth well **216**, a third contact region **214**, a second relaxation region **212**, a third depletion forming region **230**, a fourth contact region **222**, and a fourth depletion forming region **200**. The third well **210**, the fourth well **220**, the fifth well **216**, the third contact region **214**, the second relaxation region **212**, the third depletion forming region **230**, the fourth contact region **222**, and the fourth depletion forming region **200** may be regions formed by doping a semiconductor substrate (e.g., a p-type or n-type silicon (Si) substrate). For example, the semiconductor substrate may include a p-type or n-type epitaxial layer. The third well **210** may be formed by using an epitaxial growth process.

(95) The third well **210**, the fourth well **220**, the third contact region **214**, the second relaxation region **212**, the third depletion forming region **230**, the fourth contact region **222**, and the fourth depletion forming region **200** may be respectively substantially the same as the first well **110**, the second well **120**, the first contact region **114**, the first relaxation region **112**, the first depletion forming region **130**, the second contact region **122**, and the second depletion forming region **100** of FIG. **1**.

(96) The fifth well **216** may be located or provided between the third well **210** and the fourth well **220**. The fifth well **216** may be located or provided opposite to the third depletion forming region **230** with the fourth well **220** therebetween. For example, the fifth well **216** may be provided on a

bottom surface of the fourth well **220**. The fifth well **216** may have the same conductivity type as that of the third well **210**. The fifth well **216** may have the first conductivity type. For example, a doping concentration of the fifth well **216** may range from about $1 \times 10^{14} \text{ cm}^{-3}$ to about $1 \times 10^{18} \text{ cm}^{-3}$. A second main depletion region **32** may be formed in a portion adjacent to an interface between the fifth well **216** and the fourth well **220**. When a reverse bias is applied to the single-photon detection device **5**, a strong electric field may be applied to the second main depletion region **32**. For example, a maximum intensity of the electric field may range from about $1 \times 10^5 \text{ V/cm}$ to about $1 \times 10^6 \text{ V/cm}$. Because electrons may be multiplied by an electric field of the second main depletion region **32**, the second main depletion region **32** may be referred to as a multiplication region.

(97) A third sub-depletion region **12** may be formed in a portion adjacent to an interface between the fourth well **220** and the third depletion forming region **230**. The third sub-depletion region **12** may reduce or substantially prevent electrons or holes other than electron-hole pairs generated by photons in the single-photon detection device **5** from being provided to the second main depletion region **32**. For example, electrons or holes other than electron-hole pairs generated by photons in the single-photon detection device **5** may be generated due to surface defects of the single-photon detection device **5** adjacent to the third sub-depletion region **12**. The third sub-depletion region **12** may reduce or substantially prevent the electrons or holes due to the surface defects of the single-photon detection device **5** from moving to the second main depletion region **32**.

(98) A fourth sub-depletion region **22** may be formed in a portion adjacent to an interface between the fourth depletion forming region **200** and the third well **210**. The fourth sub-depletion region **22** may reduce or substantially prevent electrons or holes other than electron-hole pairs generated by photons in the single-photon detection device **5** from being provided to the second main depletion region **32**. For example, electrons or holes other than electron-hole pairs generated by photons in the single-photon detection device **5** may be generated due to surface defects of the single-photon detection device **5** adjacent to the fourth sub-depletion region **22**. The fourth sub-depletion region **22** may reduce or substantially prevent the electrons or holes due to the surface defects of the single-photon detection device **5** from moving to the second main depletion region **32**.

(99) The single-photon detection device **5** may have surface defects. Electrons or holes may be generated due to the surface defects of the single-photon detection device **5**. When the electrons or holes generated due to the surface defects move to the second main depletion region **32**, the electrons or holes may be multiplied and may generate a noise signal. That is, even when a photon is not provided to the single-photon detection device **5**, the single-photon detection device **5** may generate a noise signal.

(100) The third sub-depletion region **12** and the fourth sub-depletion region **22** of the disclosure may reduce or substantially prevent electrons or holes generated due to surface defects of the single-photon detection device **5** from moving to the second main depletion region **32**.

Accordingly, the single-photon detection device **5** having low noise may be provided. Both or only one of the third sub-depletion region **12** and the fourth sub-depletion region **22** may be implemented.

(101) FIG. **16** is a plan view illustrating the single-photon detection device of FIG. **15**, according to an embodiment.

(102) Referring to FIG. **16**, the single-photon detection device **5** may have a circular shape. In detail, the third depletion forming region **230** may have a circular shape, and the fourth contact region **222**, the fourth well **220**, the third well **210**, and the third contact region **214** may each have a circular annular shape. The fourth contact region **222**, the fourth well **220**, the third well **210**, and the third contact region **214** may surround the third depletion forming region **230**. For example, the third depletion forming region **230**, the fourth contact region **222**, the fourth well **220**, the third well **210**, and the third contact region **214** may have the same center. The fourth contact region **222**, the fourth well **220**, the third well **210**, and the third contact region **214** may be sequentially

arranged away from the third depletion forming region **230**.

(103) Although the single-photon detection device **5** has a circular shape, the disclosure is not limited thereto. In another example, as shown in FIGS. **3** through **8**, the single-photon detection device **5** may have a square shape, an octagonal shape, a square shape with round corners, a rectangular shape, a rectangular shape with round corners, or an elliptical shape.

(104) FIG. **17** is a cross-sectional view illustrating a single-photon detection device, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **15** will be omitted.

(105) Referring to FIG. **17**, a single-photon detection device **6** may be provided. The single-photon detection device **6** may include a SPAD. The single-photon detection device **6** may include the third well **210**, the fourth well **220**, the fifth well **216**, the third contact region **214**, the second relaxation region **212**, the third depletion forming region **230**, and the fourth contact region **222**. The third well **210** may be formed by using an epitaxial growth process.

(106) The single-photon detection device **6** may not include the fourth depletion forming region **200** (see FIG. **15**), unlike in FIG. **15**. Accordingly, the single-photon detection device **6** may not include the fourth sub-depletion region **22**.

(107) The third sub-depletion region **12** of the disclosure may reduce or substantially prevent electrons or holes generated due to defects of a surface of the single-photon detection device **6** adjacent to the third sub-depletion region **12** from moving to the second main depletion region **32**. Accordingly, the single-photon detection device **6** having low noise may be provided.

(108) FIG. **18** is a cross-sectional view illustrating a single-photon detector, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **1** will be omitted.

(109) Referring to FIG. **18**, a single-photon detector SE**1** may be provided. The single-photon detector SE**1** may be an image sensor using a backside illumination (BSI) method. The single-photon detector SE**1** may include a single-photon detection device **1000**, a connection layer **2000**, a control layer **2200**, and a lens unit **3000**. The single-photon detection device **1000** may be substantially the same as the single-photon detection device **1** of FIG. **1**. However, this is merely an example. In another example, the single-photon detection device **1000** may be any one of the single-photon detection devices **2**, **3**, **4**, **5**, and **6** described above or single-photon detection devices **7**, **7-1**, **7-2**, **8**, **8-1**, **8-2**, and **9** described below. The single-photon detection device **1000** may include a SPAD. For convenience of explanation, the single-photon detection device **1** of FIG. **1** is overturned and illustrated as the single-photon detection device **1000**.

(110) The connection layer **2000** may include an insulating layer **2002** and wirings **2100**. For example, the insulating layer **2002** may include silicon oxide (e.g., SiO₂), silicon nitride (e.g., SiN), silicon oxynitride (e.g., SiON), or a combination thereof.

(111) The wirings **2100** may be connected to the first contact region **114** and the second contact region **122**, and may apply a voltage. The wirings **2100** may electrically connect the second contact region **122** to the first depletion forming region **130**, or may apply the same voltage. In an example, the wirings **2100** may be widely formed over the single-photon detection device **1000** and may improve efficiency by reflecting light. The wirings **2100** may connect the first contact region **114** and/or the second contact region **122** to a circuit **2202**.

(112) The control layer **2200** may include the circuit **2202**. For example, the control layer **2200** may be a chip on which the circuit **2202** is formed. Although the circuit **2202** is shown as one block, it does not mean that the circuit **2202** includes only one electronic component (e.g., one transistor). The circuit **2202** may include various electronic components when necessary. When the single-photon detection device **1000** includes a SPAD, the circuit **2202** may include a quenching resistor or circuit and a readout circuit. The quenching circuit may stop an avalanche effect, and may enable the SPAD to detect another photon. Other pixel circuits may include a reset or recharge circuit, a memory, an amplification circuit, a counter, and a gate circuit, and may transmit signal

current to the single-photon detection device **1000** or may receive signal current from the single-photon detection device **1000**. Although the circuit **2202** is provided in the control layer **2200**, the disclosure is not limited thereto. In another example, the circuit **2202** may be located on a semiconductor substrate on which the single-photon detection device **1000** is formed.

(113) The lens unit **3000** may focus incident light and may transmit the light to the single-photon detection device **1000**. For example, the lens unit **3000** may include a microlens and a Fresnel lens. In an example, a central axis of the lens unit **3000** may be aligned with a central axis of the single-photon detection device **1000**. The central axis of the lens unit **3000** and the central axis of the single-photon detection device **1000** may be imaginary axes that respectively pass through the center of the lens unit **3000** and the center of the single-photon detection device **1000** and are parallel to a direction in which the single-photon detection device **1000** and the lens unit **3000** is stacked. In an example, the central axis of the lens unit **3000** may be misaligned with the central axis of the single-photon detection device **1000**. When the circuit **2202** is located on the semiconductor substrate on which the single-photon detection device **1000** is formed, assuming that a pixel isolation structure or a separation film (not shown) is used, the lens unit **3000** may be implemented to be larger than the single-photon detection device **1000**. In an embodiment, the single-photon detector SE1 may further include a color filter, a bandpass filter, an anti-reflection coating, a two-dimensional (2D) nanomaterial, an organic material, etc. For example, a color filter, a bandpass filter, an anti-reflection coating, a two-dimensional (2D) nanomaterial, an organic material, etc. may be formed on the single-photon detection device **1000**.

(114) According to the disclosure, the single-photon detector SE1 having low noise may be provided.

(115) FIG. **19** is a cross-sectional view illustrating a single-photon detector, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **18** will be omitted.

(116) Referring to FIG. **19**, a single-photon detector SE2 may be provided. The single-photon detector SE2 may be an image sensor using a frontside illumination (FSI) method. The single-photon detector SE2 may include the single-photon detection device **1000**, the connection device **2000**, a separation film SL, the circuit **2202**, and the lens unit **3000**. The single-photon detection device **1000** may be substantially the same as the single-photon detection device **1000** of FIG. **18**.

(117) The separation film SL may surround the single-photon detection device **1000**. For example, the separation film SL may include silicon oxide, silicon nitride, silicon oxynitride, polycrystalline silicon, a low-k dielectric material, a metal, or a combination thereof.

(118) The circuit **2202** may be located on a semiconductor substrate on which the single-photon detection device **1000** is formed. The circuit **2202** may be substantially the same as the circuit **2202** of FIG. **18**.

(119) The connection layer **2000** may be located or provided on the single-photon detection device **1000** and the separation film SL. The connection layer **2000** may include the insulating layer **2002** and the wirings **2100**. The insulating layer **2002** and the wirings **2100** may be respectively substantially the same as the insulating layer **2002** and the wirings **2100** of FIG. **18**.

(120) The lens unit **3000** may be located or provided on the connection layer **2000**. The lens unit **3000** may be substantially the same as the lens unit **3000** of FIG. **18**.

(121) In an embodiment, the single-photon detector SE2 may further include a color filter, a bandpass filter, an anti-reflection coating, a 2D nanomaterial, an organic material, etc. may be formed on the connection layer **2000** or the single-photon detection device **1000**.

(122) According to the disclosure, the single-photon detector SE2 having low noise may be provided.

(123) FIG. **20** is a plan view illustrating a single-photon detector array, according to an embodiment. FIG. **21** is a cross-sectional view taken along line I-I' of FIG. **20**. For brevity of explanation, the same description as that made with reference to FIG. **18** will be omitted.

(124) Referring to FIGS. **20** and **21**, a single-photon detector array SPA may be provided. The single-photon detector array SPA may include single-photon detectors SE that are two-dimensionally arranged. The single-photon detectors SE may respectively include the lens units **3000**. The single-photon detectors SE may respectively define pixels. A boundary between the pixels in FIG. **20** is marked by a dashed line. Each of the single-photon detectors SE may be substantially the same as the single-photon detector SE1 of FIG. **18**. However, this is not limited thereto, and each of the single-photon detectors SE may be substantially the same as the single-photon detector SE2 of FIG. **19**.

(125) The separation film SL may be located or provided between the single-photon detectors SE. The separation film SL may prevent crosstalk in which light incident on a pixel is detected in another pixel adjacent to the pixel. For example, the separation film SL may include silicon oxide, silicon nitride, silicon oxynitride, polycrystalline silicon, a low-k dielectric material, a metal, or a combination thereof.

(126) According to the disclosure, the single-photon detector array SPA having low noise may be provided.

(127) FIG. **22** is a cross-sectional view illustrating a single-photon detection device, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **15** will be omitted.

(128) Referring to FIG. **22**, a single-photon detection device **7** may be provided. The single-photon detection device **7** may include a SPAD. Unlike in FIG. **15**, the single-photon detection device **7** may not include the third depletion forming region **230** (see FIG. **15**) and the fourth well **220** (see FIG. **15**). The fourth contact region **222**, instead of the third depletion region, may be located or provided in a portion where the third depletion region is located or provided in FIG. **15**.

(129) The fifth well **216** may be located or provided under the fourth contact region **222**. The fifth well **216** may directly contact the fourth contact region **222**. The fifth well **216** may have a width less than that of the fourth contact region **222**. A third main depletion region **34** may be formed in a portion adjacent to an interface between the fifth well **216** and the fourth contact region **222**. When a reverse bias is applied to the single-photon detection device **7**, a strong electric field may be applied to the third main depletion region **34**. For example, a maximum intensity of the electric field may range from about $1 \times 10^{5.5}$ V/cm to about $1 \times 10^{6.6}$ V/cm. Because electrons may be multiplied by an electric field of the third main depletion region **34**, the third main depletion region **34** may be referred to as a multiplication region.

(130) The fourth sub-depletion region **22** may be formed in a portion adjacent to an interface between the fourth depletion forming region **200** and the third well **210**. The fourth sub-depletion region **22** may reduce or substantially prevent electrons or holes other than electron-hole pairs generated by photons in the single-photon detection device **7** from being provided to the third main depletion region **34**. For example, electrons or holes other than electron-hole pairs generated by photons in the single-photon detection device **7** may be generated due to surface defects of the single-photon detection device **7** adjacent to the fourth sub-depletion region **22**. The fourth sub-depletion region **22** may reduce or substantially prevent the electrons or holes due to the surface defects of the single-photon detection device **7** from moving to the third main depletion region **34**.

(131) According to the disclosure, the single-photon detection device **7** having low noise may be provided.

(132) FIG. **23** is a plan view illustrating the single-photon detection device of FIG. **22**, according to an embodiment.

(133) Referring to FIG. **23**, the single-photon detection device **7** may have a circular shape. In detail, the fourth contact region **222** may have a circular shape, and the third well **210** and the third contact region **214** may each have a circular annular shape. The third well **210** and the third contact region **214** may surround the fourth contact region **222**. For example, the fourth contact region **222**, the third well **210**, and the third contact region **214** may have the same center. The third well **210**

and the third contact region **214** may be sequentially arranged away from the fourth contact region **222**.

(134) Although the single-photon detection device **7** has a circular shape, the disclosure is not limited thereto. In another example, as shown in FIGS. **3** through **8**, the single-photon detection device **7** may have a square shape, an octagonal shape, a square shape with round corners, a rectangular shape, a rectangular shape with round corners, or an elliptical shape.

(135) FIG. **24** is a cross-sectional view illustrating a single-photon detection device, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **22** will be omitted.

(136) Referring to FIG. **24**, a single-photon detection device **8** may be provided. The single-photon detection device **8** may include a SPAD. Referring to FIG. **22**, the single-photon detection device **8** may further include a guard ring region **240**.

(137) The guard ring region **240** may be located or provided on a side surface of the fifth well **216**. The guard ring region **240** may prevent premature breakdown by reducing the concentration of an electric field at an edge of the fifth well **216**. Premature breakdown, which is a phenomenon where breakdown first occurs at an edge of the fifth well **216** before an electric field of a sufficient magnitude is applied to the third main depletion region **34**, occurs as an electric field is concentrated at the edge of the fifth well **216**. The guard ring region **240** may improve breakdown characteristics of the single-photon detection device **8**. The guard ring region **240** may surround the fifth well **216**. For example, the guard ring region **240** may have an annular shape extending along the side surface of the fifth well **216**. The guard ring region **240** may directly contact the fifth well **216**. The guard ring region **240** may be spaced apart from the third contact region **214** and the second relaxation region **212**. However, this is merely an example. In another example, the guard ring region **240** may directly contact the third contact region **214** and the second relaxation region **212**.

(138) The third well **210** may extend to a portion between the guard ring region **240** and the second relaxation region **212** and a portion between the guard ring region **240** and the third contact region **214**. For example, the portion between the guard ring region **240** and the second relaxation region **212** and the portion between the guard ring region **240** and the third contact region **214** may be filled with the third well **210**. The guard ring region **240** may have the same conductivity type as that of the fourth contact region **222**. A doping concentration of the guard ring region **240** may be lower than a doping concentration of the fourth contact region **222**. For example, a doping concentration of the guard ring region **240** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{18} \text{ cm}^{-3}$.

(139) According to the disclosure, the single-photon detection device **8** having low noise and improved breakdown characteristics may be provided.

(140) FIG. **25** is a plan view illustrating the single-photon detection device of FIG. **24**, according to an embodiment.

(141) Referring to FIG. **25**, the single-photon detection device **8** may have a circular shape. In detail, the fourth contact region **222** may have a circular shape, and the guard ring region **240**, the third well **210**, and the third contact region **214** may each have a circular annular shape. The guard ring region **240**, the third well **210**, and the third contact region **214** may surround the third depletion forming region **230**. For example, the fourth contact region **222**, the guard ring region **240**, the third well **210**, and the third contact region **214** may have the same center. The guard ring region **240**, the third well **210**, and the third contact region **214** may be sequentially arranged away from the fourth contact region **222**.

(142) Although the single-photon detection device **8** has a circular shape, the disclosure is not limited thereto. In another embodiment, as shown in FIGS. **3** through **8**, the single-photon detection device **8** may have a square shape, an octagonal shape, a square shape with round corners, a rectangular shape, a rectangular shape with round corners, or an elliptical shape.

(143) FIG. 26 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. 1 will be omitted.

(144) Referring to FIG. 26, a single-photon detection device 9 may be provided. The single-photon detection device 9 may include a SPAD. The single-photon detection device 9 may be substantially the same as the single-photon detection device 1 of FIG. 1, except for the first well 110 and a second conductivity-type region 101.

(145) A doping concentration of the first well 110 may have a gradient. For example, a doping concentration of the first well 110 may decrease away from the second depletion forming region 100.

(146) The second conductivity-type region 101 may be located or provided on the first well 110. Although the second conductivity-type region 101 is located or provided between the first relaxation region 112 and the second well 120 and between the first contact region 114 and the second well 120, the disclosure is not limited thereto. In another example, the second conductivity-type region 101 may be located or provided between the first contact region 114 and the second well 120, and may not be located or provided between the first relaxation region 112 and the second well 120.

(147) The second conductivity-type region 101 may have the second conductivity type. When a conductivity type of the first well 110 is n-type, a conductivity type of the second conductivity-type region 101 may be p-type. When a conductivity type of the first well 110 is p-type, a conductivity type of the second conductivity-type region 101 may be n-type. The second conductivity-type region 101 may substantially have the same doping concentration as that of the second depletion forming region 100.

(148) FIG. 27 is a plan view illustrating the single-photon detection device of FIG. 26, according to an embodiment.

(149) Referring to FIG. 27, the single-photon detection device 9 may have a circular shape. In detail, the first depletion forming region 130 may have a circular shape, and the second contact region 122, the second well 120, the first well 110, the second conductivity-type region 101, and the first contact region 114 may each have a circular annular shape. The second contact region 122, the second well 120, the first well 110, the second conductivity-type region 101, and the first contact region 114 may surround the first depletion forming region 130. For example, the first depletion forming region 130, the second contact region 122, the second well 120, the first well 110, the second conductivity-type region 101, and the first contact region 114 may have the same center. The second contact region 122, the second well 120, the first well 110, the second conductivity-type region 101, and the first contact region 114 may be sequentially arranged away from the first depletion forming region 130.

(150) Although the single-photon detection device 9 has a circular shape, the disclosure is not limited thereto. In another example, as shown in FIGS. 3 through 8, the single-photon detection device 9 may have a square shape, an octagonal shape, a square shape with round corners, a rectangular shape, a rectangular shape with round corners, or an elliptical shape.

(151) FIG. 28 is a cross-sectional view illustrating a single-photon detection device, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. 22 will be omitted.

(152) Referring to FIG. 28, a single-photon detection device 7-1 may be provided. The single-photon detection device 7-1 may include a SPAD. The single-photon detection device 7-1 may be substantially the same as the single-photon detection device 7 of FIG. 22, except that the single-photon detection device 7-1 includes a lower well 201 instead of the fourth depletion region 200 (see FIG. 22).

(153) The lower well 201 may be located or provided under the third well 210. The lower well 201 may be located or provided opposite to the fifth well 216 with the third well 210 therebetween. The

lower well **201** may be regions formed by doping a semiconductor substrate (e.g., a p-type or n-type silicon (Si) substrate). The lower well **201** may have the same conductivity type as that of the third well **210**. The lower well **201** may have the first conductivity type. The lower well **201** may be doped at a higher concentration than that of the third well **210**. For example, a doping concentration of the lower well **201** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{21} \text{ cm}^{-3}$.

(154) The lower well **201** may reduce a magnitude of an electric field generated under the single-photon detection device **7-1**. As a magnitude of an electric field generated under the single-photon detection device **7-1** decreases, a drift velocity of electrons or holes (hereinafter, noise carriers) generated due to defects on a bottom surface of the single-photon detection device **7-1** may decrease. As a drift velocity of noise carriers decreases, the influence of the noise carriers on the single-photon detection device **7-1** (e.g., a degree to which the noise carriers move to the third main depletion region **34**) may decrease. Accordingly, the single-photon detection device **7-1** of the disclosure may have low noise.

(155) FIG. **29** is a cross-sectional view illustrating a single-photon detection device, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **28** will be omitted.

(156) Referring to FIG. **29**, a single-photon detection device **7-2** may be provided. The single-photon detection device **7-2** may include a SPAD. The single-photon detection device **7-2** may be substantially the same as the single-photon detection device **7-1** of FIG. **28**, except that the single-photon detection device **7-2** includes a third relaxation region **212a** and a fourth relaxation region **212b** that is deeper than the third relaxation region **212a**, instead of the second relaxation region **212**.

(157) The third relaxation region **212a** may be located or provided under the third contact region **214**. The third relaxation region **212a** may directly contact the third contact region **214**. The third relaxation region **212a** may have the first conductivity type. A doping concentration of the third relaxation region **212a** may be lower than a doping concentration of the first contact region **114**, and may be higher than a doping concentration of the first well **110**. For example, a doping concentration of the third relaxation region **212a** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{19} \text{ cm}^{-3}$. In an embodiment, the third relaxation region **212a** may be substantially the same as the second relaxation region **212** (see FIG. **28**).

(158) The fourth relaxation region **212b** may be located or provided under the third relaxation region **212a**. The fourth relaxation region **212b** may contact the third relaxation region **212a**. The fourth relaxation region **212b** may have the first conductivity type. The fourth relaxation region **212b** may have a similar doping concentration range to that of the third relaxation region **212a**. For example, a doping concentration of the fourth relaxation region **212b** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{19} \text{ cm}^{-3}$.

(159) When the third relaxation region **212a** and the fourth relaxation region **212b** are used, a uniform bias voltage may be applied to the third well **210**. When a plurality of single-photon detection devices **7-2** are located or disposed adjacent to one another, crosstalk between adjacent single-photon detection devices **7-2** may be prevented by the third relaxation region **212a** and the fourth relaxation region **212b**.

(160) FIG. **30** is a cross-sectional view illustrating a single-photon detection device, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **24** will be omitted.

(161) Referring to FIG. **30**, a single-photon detection device **8-1** may be provided. The single-photon detection device **8-1** may include a SPAD. The single-photon detection device **8-1** may be substantially the same as the single-photon detection device **8** of FIG. **24**, except that the single-photon detection device **8-1** includes the lower well **201** instead of the fourth depletion region **200** (see FIG. **24**).

(162) The lower well **201** may be located or provided under the third well **210**. The lower well **201** may be located or provided opposite to the fifth well **216** with the third well **210** therebetween. The lower well **201** may be regions formed by doping a semiconductor substrate (e.g., a p-type or n-type silicon (Si) substrate). The lower well **201** may have the same conductivity type as that of the third well **210**. The lower well **201** may have the first conductivity type. The lower well **201** may be doped at a higher concentration than that of the third well **210**. For example, a doping concentration of the lower well **201** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{21} \text{ cm}^{-3}$.

(163) The lower well **201** may reduce a magnitude of an electric field generated under the single-photon detection device **8-1**. As a magnitude of an electric field generated under the single-photon detection device **8-1** decreases, a drift velocity of electrons or holes (hereinafter, noise carriers) generated due to defects on a bottom surface of the single-photon detection device **8-1** may decrease. As a drift velocity of noise carriers decreases, the influence of the noise carriers on the single-photon detection device **8-1** (e.g., a degree to which the noise carriers move to the third main depletion region **34**) may decrease. Accordingly, the single-photon detection device **8-1** of the disclosure may have low noise.

(164) FIG. **31** is a cross-sectional view illustrating a single-photon detection device, according to an embodiment. For brevity of explanation, the same description as that made with reference to FIG. **30** will be omitted.

(165) Referring to FIG. **31**, a single-photon detection device **8-2** may be provided. The single-photon detection device **8-2** may include a SPAD. The single-photon detection device **8-2** may be substantially the same as the single-photon detection device **8-1** of FIG. **30**, except that the single-photon detection device **8-2** includes the relaxation region **212a** and the relaxation region **212b** that is deeper than the third relaxation region **212a**, instead of the second relaxation region **212**.

(166) The third relaxation region **212a** may be located or provided under the third contact region **214**. The third relaxation region **212a** may directly contact the third contact region **214**. The third relaxation region **212a** may have the first conductivity type. A doping concentration of the third relaxation region **212a** may be lower than a doping concentration of the first contact region **114**, and may be higher than a doping concentration of the first well **110**. For example, a doping concentration of the third relaxation region **212a** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{19} \text{ cm}^{-3}$. In an example, the third relaxation region **212a** may be substantially the same as the second relaxation region **212** (see FIG. **30**).

(167) The fourth relaxation region **212b** may be located or provided under the third relaxation region **212a**. The fourth relaxation region **212b** may contact the third relaxation region **212a**. The fourth relaxation region **212b** may have the first conductivity type. The fourth relaxation region **212b** may have a similar doping concentration range to that of the third relaxation region **212a**. For example, a doping concentration of the fourth relaxation region **212b** may range from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{19} \text{ cm}^{-3}$.

(168) When the third relaxation region **212a** and the fourth relaxation region **212b** are used, a uniform bias voltage may be applied to the third well **210**. When a plurality of single-photon detection devices **8-2** are located or disposed adjacent to one another, crosstalk between adjacent single-photon detection devices **8-2** may be prevented by the third relaxation region **212a** and the fourth relaxation region **212b**.

(169) FIG. **32** is a block diagram for describing an electronic device, according to an embodiment.

(170) Referring to FIG. **32**, an electronic device **4000** may be provided. The electronic device **4000** may emit light to an object (not shown), and may detect light reflected from the object to the electronic device **4000**. The electronic device **4000** may include a beam steering device **4010**. The beam steering device **4010** may adjust a direction of light emitted to the outside of the electronic device **4000**. The beam steering device **4010** may be a mechanical or non-mechanical (semiconductor type) beam steering device. The electronic device **4000** may include a light source

unit in the beam steering device **4010**, or may include a light source unit located outside the beam steering device **4010**. The beam steering device **4010** may be a light-emitting device using a scanning method. However, the light-emitting device of the electronic device **4000** is not limited to the beam steering device **4010**. In another example, the electronic device **4000** may include a light-emitting device using a flash method, instead of or along with the beam steering device **4010**. The light-emitting device using the flash method may emit light to an area including all fields of view at once without a scanning process.

(171) Light steered by the beam steering device **4010** may be reflected by the object to the electronic device **4000**. The electronic device **4000** may include a detection unit **4020** for detecting light reflected by the object. The detection unit **4020** may include a plurality of light detection elements, and may further include other optical members. The plurality of light detection elements may include any one of the single-photon detection devices **1**, **2**, **3**, **4**, **5**, **6**, **7**, **7-1**, **7-2**, **8**, **8-1**, **8-2**, and **9** described above. Also, the electronic device **4000** may further include a circuit unit **4030** connected to at least one of the beam steering device **4010** and the detection unit **4020**. The circuit unit **4030** may include a calculation unit that obtains and calculates data, and may further include a driver and a controller. Also, the circuit unit **4030** may further include a power supply unit and a memory.

(172) Although the electronic device **4000** includes the beam steering device **4010** and the detection unit **4020** in one device, the beam steering device **4010** and the detection unit **4020** may not be provided in one device but may be provided in separate devices. Also, the circuit unit **4030** may be connected to the beam steering device **4010** or the detection unit **4020** through wireless communication rather than wired communication.

(173) The electronic device **4000** according to the above embodiment may be applied to various electronic devices. For example, the electronic device **4000** may be applied to a light detection and ranging (LiDAR) device. The LiDAR device may be a phase-shift type or time-of-flight (TOF) type device. Also, any of the single-photon detection devices **1**, **2**, **3**, **4**, **5**, **6**, **7**, **7-1**, **7-2**, **8**, **8-1**, **8-2**, and **9** or the electronic device **4000** including the same according to an embodiment may be mounted on an electric device such as a smartphone, a wearable device (e.g., augmented reality (AR) and virtual reality (VR) glasses), an Internet of things (IoT) device, a home appliance, a tablet personal computer (PC), a personal digital assistant (PDA), a portable multimedia player (PMP), a navigation device, a drone, a robot, a self-driving vehicle, an autonomous vehicle, or an advanced driver-assistance system (ADAS).

(174) FIGS. **33** and **34** are conceptual views illustrating a case where a LiDAR device is applied to a vehicle, according to an embodiment.

(175) Referring to FIGS. **33** and **34**, a LiDAR device **5010** may be applied to a vehicle **5000**, and information about an object **6000** may be obtained by using the LiDAR device **5010**. The vehicle **5000** may be an autonomous vehicle. The LiDAR device **5010** may detect a solid body or a person, that is, the object **6000**, in a direction in which the vehicle **5000** travels. The LiDAR device **5010** may measure a distance to the object **6000**, by using information such as a time difference between a transmitted signal and a detected signal. The LiDAR device **5010** may obtain information about a near object **6010** and a far object **6020** within a scan range. The LiDAR device **5010** may include the electronic device **4000** of FIG. **32**.

(176) Although the LiDAR device **5010** is located or disposed on a front portion of the vehicle **5000** and detects the object **6000** in a direction in which the vehicle **5000** travels, the disclosure is not limited thereto. In another example, the LiDAR device **5010** may be located or disposed at a plurality of locations on the vehicle **5000** to detect all objects **6000** around the vehicle **5000**. For example, four LiDAR devices **5010** may be located or disposed on a front portion, a rear portion, and left and right portions of the vehicle **5000**. In another example, the LiDAR device **5010** may be located or disposed on a roof of the vehicle **5000**, and may rotate and detect all objects **6000** around the vehicle **5000**.

(177) According to the disclosure, a photo-detection device having low noise may be provided.
(178) According to the disclosure, a photodetector having low noise may be provided.
(179) However, effects of the disclosure are not limited thereto.
(180) It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the following claims.

Claims

1. A single-photon detection device comprising: a first well having a first conductivity type; a second well provided on the first well and having a second conductivity type that is different from the first conductivity type; a first depletion forming region provided on the second well and having the first conductivity type; a main depletion region provided between the first well and the second well as an avalanche multiplication region; and a first sub-depletion region provided between the second well and the first depletion forming region, wherein the first well and the first depletion forming region are spaced apart from each other so that the first depletion forming region does not directly contact the first well.
2. The single-photon detection device of claim 1, further comprising: a second depletion forming region provided opposite to the second well with the first well therebetween; and a second sub-depletion region provided between the second depletion forming region and the first well, wherein the second depletion forming region has the second conductivity type.
3. The single-photon detection device of claim 1, further comprising a first contact region electrically connected to the first well, wherein the first contact region is spaced apart from the second well and has the first conductivity type, wherein a doping concentration of the first contact region is higher than a doping concentration of the first well.
4. The single-photon detection device of claim 3, further comprising a relaxation region provided between the first contact region and the first well, wherein the relaxation region has the first conductivity type, wherein a doping concentration of the relaxation region is lower than a doping concentration of the first contact region.
5. The single-photon detection device of claim 1, further comprising a second contact region electrically connected to the second well, wherein the second contact region has the second conductivity type, wherein a doping concentration of the second contact region is higher than a doping concentration of the second well.
6. The single-photon detection device of claim 5, wherein the second contact region and the first depletion forming region contact each other.
7. The single-photon detection device of claim 5, wherein the second contact region and the first depletion forming region are spaced apart from each other.
8. The single-photon detection device of claim 1, further comprising a guard ring region provided on a side surface of the second well, wherein the guard ring region has the second conductivity type.
9. The single-photon detection device of claim 8, wherein the guard ring region surrounds the second well.
10. The single-photon detection device of claim 1, wherein the first well extends from a bottom surface of the second well to a side surface of the second well.
11. The single-photon detection device of claim 1, further comprising a third well surrounding the first well and the second well, wherein the third well has the first conductivity type.

12. The single-photon detection device of claim 11, further comprising: a second depletion forming region provided opposite to the first well with the third well therebetween; and a third sub-depletion region provided between the second depletion forming region and the third well, wherein the second depletion forming region has the second conductivity type.
 13. The single-photon detection device of claim 11, further comprising a third contact region provided on the third well and spaced apart from the second well, wherein the third contact region has the first conductivity type.
 14. The single-photon detection device of claim 11, wherein a doping concentration of the third well is lower than a doping concentration of the first well.
 15. The single-photon detection device of claim 11, wherein the third well contacts a side surface of the second well.
 16. A single-photon detector comprising: a single-photon detection device configured to generate an electrical signal by receiving light; a control layer configured to process the electrical signal; and a lens unit configured to receive the light from outside and focus the light on the single-photon detection device, wherein the single-photon detection device comprises a first well having a first conductivity type, a second well provided on the first well and having a second conductivity type that is different from the first conductivity type, a first depletion forming region provided on the second well and having the first conductivity type, a main depletion region provided between the first well and the second well as an avalanche multiplication region, and a first sub-depletion region provided between the second well and the first depletion forming region, wherein the first well and the first depletion forming region are spaced apart from each other so that the first depletion forming region does not directly contact the first well.
 17. The single-photon detector of claim 16, wherein the control layer is disposed between the single-photon detection device and the lens unit.
 18. The single-photon detector of claim 16, wherein the single-photon detection device is disposed between the control layer and the lens unit.
 19. The single-photon detector of claim 16, wherein the first depletion forming region is disposed closer to the control layer than the lens unit.
 20. The single-photon detector of claim 16, wherein the control layer comprises: an insulating layer; wirings; and circuits, wherein the wirings electrically connect the first well and the second well to the circuits, and the circuits comprise a quenching circuit and a readout circuit.
 21. The single-photon detector of claim 20, wherein the circuits further comprise at least one of a recharge circuit, a memory, an amplification circuit, a counter, and a gate circuit.
 22. The single-photon detector of claim 16, wherein the single-photon detector comprises at least one of a color filter, a bandpass filter, an anti-reflection coating, a two-dimensional (2D) nanomaterial, and an organic material provided on the single-photon detection device.
 23. A single-photon detector array comprising: a plurality of single-photon detection devices configured to generate an electrical signal by receiving light; a control layer configured to process the electrical signal; and a plurality of lenses configured to receive the light from outside and respectively focus the light on the plurality of single-photon detection devices, wherein each of the plurality of single-photon detection devices comprises a first well having a first conductivity type, a second well provided on the first well and having a second conductivity type that is different from the first conductivity type, a first depletion forming region provided on the second well and having the first conductivity type, a main depletion region provided between the first well and the second well as an avalanche multiplication region, and a first sub-depletion region provided between the second well and the first depletion forming region, wherein the first well and the first depletion forming region are spaced apart from each other so that the first depletion forming region does not directly contact the first well.
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